# Multiple Linear and Nonlinear Regression

AGRON 5130 - Unit 11 Assignment

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## Introduction

The goal of this assignment is to help you practice the full workflow from Module 11:

- interpreting partial effects in multiple regression
- diagnosing multicollinearity and unstable coefficients
- comparing full vs reduced models
- fitting and interpreting a nonlinear growth model
- connecting model choice to biological reasoning

You will complete two parts:

- **Part 1:** multiple linear regression using soybean yield data
- **Part 2:** nonlinear modeling using tomato growth data

Submit a knitted PDF/HTML generated from your R Markdown file to Canvas.

**Total: 15 points**

## Part 1: Multiple Linear Regression (Thompson Soy)

```r
library(ggplot2)
library(car)
```

```
## Loading required package: carData
```

```r
thompson_soy <- read.csv("data/thompson_soy.csv")
head(thompson_soy)
```

```
##   yield rain6 rain7 temp6 temp7
## 1    17  3.36  1.01  71.9  79.1
## 2    18  3.19  2.97  75.5  79.3
## 3    20  4.08  3.32  74.0  77.6
## 4    15  1.48  2.41  78.2  78.1
## 5    19  3.01  3.24  79.0  81.7
## 6    18  5.99  3.33  68.2  79.0
```

For this dataset:

- `yield` = soybean yield
- `rain6` = total June precipitation
- `rain7` = total July precipitation
- `temp6` = mean June temperature
- `temp7` = mean July temperature

The goal is to ask which of these weather variables are associated with yield **after accounting for the others**.

# Question 1

**1 point**

Fit a multiple linear regression model of `yield` as a function of all four weather variables:

- `rain6`
- `rain7`
- `temp6`
- `temp7`

# Question 2

**1 point**

Report model coefficients.

Write 2-3 sentences interpreting:

1. one precipitation coefficient as a partial effect
2. one temperature coefficient as a partial effect

Use language like: "Holding the other predictors constant…"

# Question 3

**1 point**

Report model fit statistics from `summary()`:

# Question 4

**1 point**

Create a scatterplot matrix of model variables.

Then name one pair of predictors that appears strongly related.

# Question 5

**1 point**

Create a correlation matrix for the predictor variables only.

# Question 6

**1 point**

Describe and interpret the correlations between variables. Are there any varaibles you would be concerned about including in the model? Why?

# Question 7

**1 point**

Compute VIF values for the full model.

Then answer:

1. Which predictor has the largest VIF?
2. Is multicollinearity likely to affect interpretation?

# Question 8

**1 point**

Create a reduced model by dropping one predictor you believe is least useful or most redundant.

Compare full vs reduced model using:

1. ANOVA (`anova(reduced, full)`)
2. AIC (`AIC(reduced, full)`)

In 3-4 sentences, justify which model you would keep for interpretation.

# Question 9

**1 point**

Check assumptions for your **selected** model using:

1. residuals vs fitted plot (`which = 1`)
2. normal Q-Q plot (`which = 2`)

Write 2-3 sentences on whether the linear model is adequate.

# Part 2: Nonlinear Regression (Tomato Growth)

```
tomato <- read.csv("data/tomato_days_volume.csv")
head(tomato)
```

```
##   days fruit_volume
## 1    7            5
## 2   14           40
## 3   21           75
## 4   28          110
## 5   35          157
## 6   42          180
```

## Question 10

### 1 point

Plot `fruit_volume` as a function of `days`.

In one sentence, describe the shape you see.

## Question 11

### 2 points

Fit and summarize a logistic nonlinear model:

`fruit_volume ~ SSlogis(days, Asym, xmid, scal)`

Interpret each parameter in biological terms:

1. `Asym`
2. `xmid`

## Question 12

### 1 point

Create a prediction grid for `days` from 7 to 49 by 1, then predict `fruit_volume` on that grid.

## Question 13

### 1 point

Add the fitted logistic curve to your original tomato scatterplot.

# Question 14

**1 point**

Fit a polynomial model to the same tomato data:

```
lm(fruit_volume ~ days + I(days^2), data = tomato)
```

Then, in 3-4 sentences, compare polynomial vs logistic for this dataset using:

1. biological interpretability
2. plausibility of curve shape
3. model communication to an agronomy audience